

Language Specific

Interview Cracker question bank

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Java Interview Questions

[Easy] Q. What is the difference between JDK, JRE and JVM?

Answer. **JVM** executes bytecode and is the thing that makes Java portable - one `.class` file runs anywhere a JVM exists. **JRE** is the JVM plus the standard class libraries: enough to run Java. **JDK** is the JRE plus development tools (`javac`, `jdb`, `jar`): enough to build Java.

"Write once, run anywhere" is a property of bytecode plus a per-platform JVM, not of the source language.

[Easy] Q. Primitive vs reference types?

Answer. Primitives (`int`, `double`, `boolean`, `char`, ...) hold values directly, live on the stack when local, cannot be null, and have no methods. Reference types hold a pointer to a heap object, can be null, and carry methods.

The consequence that produces bugs:

```
String a = new String("hi");
String b = new String("hi");
a == b           // false - different objects
a.equals(b)      // true  - same characters
```

`==` compares references for objects, values for primitives. **Always use `.equals()` for objects.**

Autoboxing makes this worse:

```
Integer x = 127, y = 127;
x == y;           // true  - cached (-128..127)
Integer p = 128, q = 128;
p == q;           // false - outside the cache
```

Same code, different answer depending on the value. Never compare boxed types with `==`.

[Hard] Q. Explain the equals/hashCode contract.

Answer. If `a.equals(b)` then `a.hashCode() == b.hashCode()`, **always**. The reverse is not required - equal hash codes with unequal objects is just a collision.

Override one and you must override the other. Break it and hash-based collections fail silently:

```
Map<Point, String> map = new HashMap<>();
map.put(new Point(1, 2), "origin-ish");
map.get(new Point(1, 2)); // null, if hashCode() was not overridden
```

The lookup hashes to a different bucket and never finds the entry.

The second half of the contract: **a key's hash must not change while it is in a collection**. Mutating a field used by `hashCode` strands the entry - present but unreachable, and `remove()` will not find it either. Prefer immutable keys; `record` types give you correct `equals/hashCode` for free.

[Medium] Q. Compare the collections.

Answer.

Interface	Implementation	Ordering	Notes
List	ArrayList	insertion	O(1) index, O(n) middle insert. The default.

	LinkedList	insertion	O(1) ends, poor locality - rarely the right choice
Set	HashSet	none	O(1) average
	LinkedHashSet	insertion	O(1), remembers order
	TreeSet	sorted	O(log n), needs Comparable/Comparator
Map	HashMap	none	O(1) average; chains become trees past 8 entries
	LinkedHashMap	insertion or access	access-order mode gives you an LRU cache
	TreeMap	sorted	O(log n), range queries

`ArrayList` unless you need ordering or sorting. `LinkedList` is almost never the answer despite what its Big-O table suggests - cache locality wins.

[Medium] Q. Checked vs unchecked exceptions?

Answer. Checked (extend `Exception`) must be declared or caught - the compiler enforces it. Intended for recoverable conditions: `IOException`, `SQLException`.

Unchecked (extend `RuntimeException`) need no declaration. Intended for programming errors: `NullPointerException`, `IllegalArgumentException`, `IndexOutOfBoundsException`.

`Error` (e.g. `OutOfMemoryError`, `StackOverflowError`) signals a JVM-level problem you should not catch.

The design debate is worth knowing: checked exceptions were meant to force handling, but in practice they encourage `catch (Exception e) {}` and leak implementation details through signatures. Later languages (C#, Kotlin, Scala) deliberately omitted them. Worked example:

`examples/java/exceptions/exception_handling_basics.java`.

[Hard] Q. What is type erasure?

Answer. Generics are a **compile-time** feature. The compiler checks types, then erases them - `List<String>` becomes plain `List` in the bytecode, with casts inserted automatically. This was done for backwards compatibility with pre-generics code.

The consequences you can be asked to explain:

```
List<String> a = new ArrayList<>();
List<Integer> b = new ArrayList<>();
a.getClass() == b.getClass();      // true - both are just ArrayList

// Illegal, because the type is not available at runtime:
// if (obj instanceof List<String>) { }
// T[] array = new T[10];
```

You also cannot overload on `List<String>` vs `List<Integer>` - after erasure both methods have identical signatures. Contrast with C#, where generics are reified and the runtime does know the type argument.

[Medium] Q. How does garbage collection work?

Answer. The JVM reclaims objects no longer **reachable** from GC roots (stack references, statics, JNI references). Note it is reachability, not reference counting - so cycles are collected correctly.

The design rests on the **generational hypothesis**: most objects die young. So the heap is split:

- **Young generation** (Eden + two survivor spaces) - collected frequently with
- **Old generation** - objects that survived several collections; collected

Modern collectors (G1, ZGC, Shenandoah) do most work concurrently to keep pause times low - ZGC targets sub-millisecond pauses on large heaps.

`System.gc()` is a suggestion the JVM may ignore. You cannot force collection, and calling it is almost always a mistake. Memory leaks in Java are real but take a different form: objects still referenced but never used - a static collection that only ever grows, an unremoved listener, an unclosed resource.

[Medium] Q. What does the Stream API give you?

Answer. A declarative pipeline over a sequence, with lazy intermediate operations and one terminal operation that triggers execution.

```
List<String> names = people.stream()
    .filter(p -> p.age() >= 18)           // intermediate, lazy
    .map(Person::name)                   // intermediate, lazy
    .sorted()
    .toList();                          // terminal, runs the pipeline
```

Nothing executes until the terminal operation, which allows fusion and short-circuiting (`findFirst` stops early).

Two cautions worth raising unprompted: a stream is **single-use** - reusing one throws

`IllegalStateException`; and `parallelStream()` is not free speed. It uses the common `ForkJoinPool`, so it helps only for large datasets with CPU-bound, independent, side-effect-free work, and can be slower otherwise. Worked example: `examples/java/streams/group_and_collect_with_streams.java`.

[Medium] Q. `String`, `StringBuilder`, `StringBuffer`?

Answer. `String` is **immutable** - every concatenation allocates a new object, so building in a loop is $O(n^2)$.

`StringBuilder` is a mutable buffer, not thread-safe, and is what you want. `StringBuffer` is the synchronised version: slower, and almost never needed since a builder is normally local to one thread.

Note the compiler rewrites simple + concatenation into builder calls, so `"a" + b + "c"` is fine - it is concatenation **inside a loop** that is the problem. Worked example:

`examples/java/strings/string_immutability_and_stringbuilder.java`.

JavaScript Interview Questions

[Easy] Q. `var`, `let` and `const`?

Answer.

	Scope	Hoisting	Reassign
<code>var</code>	function	hoisted, initialised undefined	yes
<code>let</code>	block	hoisted into the temporal dead zone	yes
<code>const</code>	block	same as <code>let</code>	no

`const` prevents **rebinding the name**, not mutation of the value:

```
const user = { name: "Sara" };
user.name = "Ali";           // fine - the object is mutable
user = {};                   // TypeError - cannot rebound
```

Use `const` by default, `let` when you must reassign, and `var` never.

[Medium] Q. What is a closure?

Answer. A function that retains access to the variables of the scope it was created in, even after that scope has returned.

```
function counter() {
  let count = 0;           // stays alive via the closure
  return {
    increment: () => ++count,
    value: () => count,
  };
}
const c = counter();
c.increment(); c.increment();
c.value();           // 2
```

`count` is genuinely private - unreachable except through the returned functions. This is the basis of the module pattern, memoisation, and every callback that "remembers" something.

The classic interview trap:

```
for (var i = 0; i < 3; i++) setTimeout(() => console.log(i)); // 3, 3, 3
for (let i = 0; i < 3; i++) setTimeout(() => console.log(i)); // 0, 1, 2
```

`var` is function-scoped, so all three callbacks close over **one** variable, which is 3 by the time they run. `let` creates a fresh binding per iteration. Worked example: examples/javascript/basics/closures.js.

[Hard] Q. Explain the event loop.

Answer. JavaScript is single-threaded but non-blocking. The runtime has:

- a **call stack** - what is executing now;
- **Web APIs / host APIs** - timers, network, file I/O, running outside the
- a **macrotask queue** - `setTimeout`, I/O callbacks, UI events;
- a **microtask queue** - promise callbacks, `queueMicrotask`.

The loop: when the stack empties, **drain the entire microtask queue**, then take **one** macrotask, then drain microtasks again. Microtasks always have priority.

```
console.log("1");
setTimeout(() => console.log("2"), 0); // macrotask
Promise.resolve().then(() => console.log("3")); // microtask
console.log("4");
// 1, 4, 3, 2
```

2 comes last despite a 0 ms delay, because the microtask queue is drained first. The follow-up: an infinite chain of microtasks **starves** the macrotask queue and freezes the page - microtask priority is not free. Worked example: `examples/javascript/async/promises_and_event_loop.js`.

[Medium] Q. How does `this` get its value?

Answer. In a normal function `this` is determined by **how it is called**, not where it is defined. Four rules, in precedence order:

- 1 **new** - `this` is the new object.
- 2 **Explicit** - `call`, `apply`, `bind` set it.
- 3 **Implicit** - `obj.method()` sets `this` to `obj`.
- 4 **Default** - `undefined` in strict mode, `globalThis` otherwise.

Arrow functions ignore all four: they have **no** `this` **of their own** and inherit it lexically from the enclosing scope. That is the fix for the classic bug:

```
class Timer {
  constructor() { this.seconds = 0; }
  startBroken() { setInterval(function () { this.seconds++; }, 1000); } // wrong `this`
  start()       { setInterval(() => { this.seconds++; }, 1000); }      // correct
}
```

Worked example: `examples/javascript/basics/this_and_binding.js`.

[Medium] Q. `==` vs `===`?

Answer. `===` compares type and value with no conversion. `==` coerces first, using rules that are genuinely surprising:

```
0 == ""; // true
0 == "0"; // true
"" == "0"; // false <- not transitive!
null == undefined // true
null == 0 // false
NaN == NaN // false
```

Always use `===`. The one accepted exception is `x == null`, which checks for `null` or `undefined` in a single test.

For `NaN`, use `Number.isNaN(x)`. `NaN !== NaN` is required by IEEE 754 and is also why `[NaN].includes(NaN)` is true (it uses `SameValueZero`) while `[NaN].indexOf(NaN)` is `-1` (it uses `===`).

[Medium] Q. What is prototypal inheritance?

Answer. Every object has a hidden link to another object, its **prototype**. A property miss walks up that chain until it is found or the chain ends at `null`.

```
const animal = { speak() { return `${this.name} makes a sound`; } };
const dog = Object.create(animal);
dog.name = "Rex";
dog.speak(); // found on the prototype
```

`class` syntax is sugar over this - there are no classical classes underneath, just constructor functions and prototype objects. That is why you can add a method to every existing instance at runtime by assigning to the prototype, which classical inheritance does not allow.

[Medium] Q. Promise, async/await, and how do you run things in parallel?

Answer. A promise represents a value that is not ready yet, in one of three states: pending, fulfilled, rejected. `async/await` is syntax over promises - an `async` function always returns one, and `await` suspends until it settles.

The mistake that matters:

```
// Sequential - 3 seconds, and usually a bug
const a = await fetchA();
const b = await fetchB();

// Parallel - 1 second
const [a, b] = await Promise.all([fetchA(), fetchB()]);
```

If the calls are independent, awaiting them one at a time wastes the whole latency of each. Know the four combinators:

- `Promise.all` - all succeed, or reject on the **first** failure.
- `Promise.allSettled` - wait for all, never rejects; inspect each result.
- `Promise.race` - first to **settle**, success or failure (used for timeouts).
- `Promise.any` - first to **succeed**.

Worked example: `examples/typescript/async/async_await_parallel_vs_sequential.ts`.

[Medium] Q. Debounce vs throttle?

Answer. Both limit how often a function runs, in opposite ways.

- **Debounce** - wait until the events stop, then run once. Search-as-you-type,
- **Throttle** - run at most once per interval while events continue. Scroll

"Fire after the user stops typing" is debounce; "fire at most every 200 ms while scrolling" is throttle. Worked example: `examples/javascript/utilities/debounce_with_cancellation.js`.

Python Interview Questions

[Easy] Q. Mutable vs immutable - and why does it matter?

Answer. **Immutable:** int, float, str, tuple, frozenset, bytes. **Mutable:** list, dict, set, bytearray, most custom classes.

It decides three things: whether a value can be a dict key (hashable, which requires immutable), whether passing it to a function can change the caller's object, and whether it is safe to share between threads.

The classic bug it causes:

```
def append_to(item, target=[]):      # DANGER: evaluated once, at def time
    target.append(item)
    return target

append_to(1)    # [1]
append_to(2)    # [1, 2]  <- the same list, still there
```

Default arguments are evaluated **once**, when the function is defined. Use `None` as the sentinel:

```
def append_to(item, target=None):
    if target is None:
        target = []
    target.append(item)
    return target
```

[Easy] Q. is vs ==?

Answer. `==` compares **values** (via `__eq__`); `is` compares **identity** - whether both names point to the same object.

```
a = [1, 2, 3]
b = [1, 2, 3]
a == b    # True  - same contents
a is b    # False - two distinct objects
```

Use `is` only for singletons: `is None`, `is True`, `is False`.

The confusing part is small-integer and string caching. `256 is 256` is `True` because CPython interns small integers (-5 to 256); `1000 is 1000` may be `False` depending on context. This is an implementation detail you must never rely on - and a favourite trick question.

[Hard] Q. What is the GIL and what does it mean for you?

Answer. The Global Interpreter Lock is a mutex ensuring only one thread executes Python bytecode at a time, even on a many-core machine. It exists because CPython's reference-counting memory management is not thread-safe, and one coarse lock was far simpler and faster for single-threaded code than locking every object.

The practical consequence:

- **CPU-bound work gets no speedup from threads.** Four threads on four cores
- **I/O-bound work does benefit**, because the GIL is released around blocking

Worth knowing this is changing: PEP 703 adds an optional free-threaded build (3.13+) with no GIL, and 3.12+ supports per-interpreter GILs. Saying "the GIL is being removed as an option, gradually" is more accurate than either "it's gone" or "Python can't do parallelism".

[Medium] Q. What is a generator and why use one?

Answer. A function containing `yield`. Calling it returns a lazy iterator that computes values on demand rather than building a list.

```
def read_large_file(path):
    with open(path) as f:
        for line in f:
            yield line.strip()          # one line in memory, not the whole file
```

Two benefits: **constant memory** regardless of input size, and the ability to represent infinite sequences. The costs: you can only iterate once, and you cannot index or take `len()`.

Use `()` instead of `[]` to get a generator expression: `sum(x*x for x in range(10**8))` never allocates the list. Worked example: `examples/python/generators/log_pipeline_with_generators.py`.

[Medium] Q. What is a decorator?

Answer. A callable that takes a function and returns a replacement, used to add behaviour without editing the original.

```
import functools, time

def timed(func):
    @functools.wraps(func)          # preserve __name__, __doc__, signature
    def wrapper(*args, **kwargs):
        start = time.perf_counter()
        try:
            return func(*args, **kwargs)
        finally:
            print(f"{func.__name__} took {time.perf_counter() - start:.4f}s")
    return wrapper

@timed
def work(): ...

@timed is exactly work = timed(work).
```

`functools.wraps` **is the detail interviewers check for**. Without it the decorated function reports the wrapper's name and loses its docstring, which breaks introspection, documentation tools and debugging. Worked example: `examples/python/decorators/retry_decorator_with_backoff.py`.

[Medium] Q. Shallow vs deep copy?

Answer. A **shallow copy** (`list(x)`, `x[:]`, `copy.copy`) makes a new outer object holding **references to the same inner objects**. A **deep copy** (`copy.deepcopy`) recursively copies everything.

```
import copy
original = [[1, 2], [3, 4]]
shallow = copy.copy(original)
shallow[0].append(99)
original          # [[1, 2, 99], [3, 4]]  <- mutated!

deep = copy.deepcopy(original)
deep[0].append(100)
original          # unchanged
```

Deep copy is slower and fails or loops on some structures - it handles cycles, but not open file handles or sockets. Prefer immutable data over copying.

[Medium] Q. What does a context manager do?

Answer. Guarantees cleanup via `__enter__` / `__exit__`, so a resource is released even if an exception is raised.

```
from contextlib import contextmanager

@contextmanager
def timer(label):
    start = time.perf_counter()
    try:
        yield
    finally:
        # runs on exception too
        print(f"{label}: {time.perf_counter() - start:.3f}s")
```

`with open(...)` is the everyday example: the file closes on any exit path. Worked example: `examples/python/context-managers/custom_context_managers.py`.

[Medium] Q. `__str__` vs `__repr__`?

Answer. `__repr__` is for **developers** - unambiguous, ideally valid Python that would recreate the object. It is what the REPL and containers show. `__str__` is for **users** - readable.

If you implement only one, implement `__repr__`: `__str__` falls back to it, but not the other way round. Worked example: `examples/python/oop/dunder_methods.py`.

[Medium] Q. What are `*args` and `**kwargs`?

Answer. `*args` collects extra positional arguments into a tuple; `**kwargs` collects extra keyword arguments into a dict. In a call, the same symbols unpack:

```
def f(a, b, c): ...
values = [1, 2, 3]
f(*values)                # unpacks positionally

config = {"a": 1, "b": 2, "c": 3}
f(**config)               # unpacks by keyword
```

This is what lets a decorator wrap any function signature.

Rust

Rust interviews test one idea above all: do you understand **ownership**, and can you reason about when the borrow checker accepts or rejects code? The language features (traits, enums, iterators) are the icing; memory safety without a garbage collector is the substance.

[Easy] Q. What is ownership, and what does it mean to "move" a value?

Answer. Every value has exactly one **owner**; when the owner goes out of scope, the value is dropped.

Assignment and function arguments **move** the value by default - the original binding is no longer usable. This is what guarantees no double-free and no use-after-free without a GC.

```
fn takes_ownership(x: String) -> u32 {
    x.len() as u32
}

fn main() {
    let s = String::from("hello");
    let n = takes_ownership(s);    // `s` is MOVED into the function
    assert_eq!(n, 5);
    // println!("{}", s);          // COMPILER ERROR: `s` was moved
}
```

Types that implement `Copy` (`ints`, `bool`, `floats`, `char`, and `fixed-size arrays` of those) are copied instead of moved, which is why `let y = x;` feels free for `i32` but not for `String`.

[Medium] Q. What are borrowing and lifetimes, and when does the borrow

checker reject your code?

Answer. A **borrow** (`&T` / `&mut T`) lets you use a value without taking ownership. The rules: any number of shared borrows, OR exactly one mutable borrow - never both at once. A **lifetime** annotation tells the compiler how long a reference may outlive the data it points to; you rarely write them explicitly, but `longest<'a>(a: &'a str, b: &'a str) -> &'a str` shows the rule that the returned reference may live no longer than the shorter-lived input.

```
fn longest<'a>(a: &'a str, b: &'a str) -> &'a str {
    if a.len() >= b.len() { a } else { b }
}

fn first_word(s: &str) -> &str {
    s.split_whitespace().next().unwrap_or("")
}

fn main() {
    let s1 = String::from("long string is long");
    let s2 = String::from("xyz");
    let result = longest(s1.as_str(), s2.as_str());
    assert_eq!(result, "long string is long");
    assert_eq!(first_word(" hello world"), "hello");
}
```

The classic rejection, and the one to be able to explain out loud:

```
fn main() {
    let result;
    {
        let s = String::from("hello");
        result = longest(s.as_str(), "world");
    }                                // `s` dropped here
    let _ = result.len();           // ERROR: `s` does not live long enough
}
```

```
}
```

`result` borrows from `s`, which dies at the closing brace - returning a reference to it would dangle. The fix is to return an owned `String` (`.to_string()`) or keep `s` alive for as long as `result` is used.

[Medium] Q. How does Rust handle errors - `Option`, `Result`, and the `?`

operator?

Answer. Failure is a **value**, not an exception. `Option<T>` is `Some(T)` or `None` for "no value"; `Result<T, E>` is `Ok(T)` or `Err(E)` for recoverable failures. The `?` operator propagates an error: on `Err` it returns early from the current function, on `Ok` it unwraps the value - replacing the nested-matching pyramid of code.

```
fn parse_int(s: &str) -> Result<i64, String> {
    s.parse::<i64>().map_err(|e| format!("not a number: {e}"))
}

fn safe_div(a: i64, b: i64) -> Result<i64, String> {
    if b == 0 { return Err("division by zero".to_string()); }
    Ok(a / b)
}

// `?` returns early on Err, unwraps on Ok:
fn process(input: &str) -> Result<i64, String> {
    let n = parse_int(input)?;
    let half = safe_div(n, 2)?;
    Ok(half)
}

fn main() {
    assert_eq!(process("10"), Ok(5));
    assert!(process("abc").is_err());
    assert_eq!(safe_div(1, 0), Err("division by zero".to_string()));
}
```

The trade-off to state: `?` makes pipelines linear but forces every function to thread a `Result` through its signature - errors are explicit at every boundary (a feature), whereas exception-based languages let failures surface only where they are caught (and silently, if never caught).

[Medium] Q. When do you reach for `Box`, `Rc`, or `RefCell`?

Answer. They solve the three limitations of the stack-ownership model:

- `Box<T>` - heap allocation / unknown-at-compile-time size /
- `Rc<T>` - shared ownership: many owners, one data. Reference
- `RefCell<T>` - interior mutability: immutable binding, mutable

```
use std::rc::Rc;
use std::cell::RefCell;

struct Node {
    value: i32,
    next: RefCell<Option<Rc<Node>>>,
}

fn push(head: &Rc<RefCell<Option<Rc<Node>>>>, value: i32) {
    let new = Rc::new(Node { value, next: RefCell::new(None) });
    (*head).borrow_mut().replace(new); // unique write access for a moment
}

fn len(head: &Rc<RefCell<Option<Rc<Node>>>>) -> usize {
```

```

    let mut n = 0;
    let mut cur = head.borrow().clone(); // clone the Rc (cheap refcount bump)
    while let Some(node) = cur {
        n += 1;
        cur = node.next.borrow().clone();
    }
    n
}

```

`Rc` outside + `RefCell` inside is the standard shape for data with multiple owners AND runtime-rearranged structure (graphs, trees you mutate in place). The cost: `borrow()/borrow_mut()` panic at runtime on rule violations, and `Rc` increments/decrements a counter on every clone/drop.

[Hard] Q. Walk through what happens when a function with an `Option`

return value is called, and why `unwrap()` is a code smell in production.

Answer. The caller **must** handle both cases - the type system makes `None` a visible branch, usually via `match`, `if let`, or `? .unwrap()` (and `expect()`) say "I guarantee this is `Some`, panic otherwise" - fine in tests and prototypes, but in production it converts a handleable absence into a crash, typically deep in a request path where the real cause (a bad input, a missing config key) is far away. The production pattern is to propagate: return a `Result/Option` upward with `?`, map the `None` into a meaningful error with `ok_or(...)`, or branch deliberately with a default only when a default is genuinely correct.

```

fn find_user(id: u64) -> Option<String> {
    (id == 1).then(|| String::from("ada"))
}

fn greet(id: u64) -> Result<String, String> {
    let user = find_user(id)
    .ok_or_else(|| format!("no user with id {id}"))?;
    Ok(format!("hello {user}"))
}

fn main() {
    assert_eq!(greet(1), Ok("hello ada".to_string()));
    assert!(greet(2).is_err());
}

```

The follow-up: `Option` and `Result` are enums with **zero overhead** in the `Some/Ok` case (a niche optimization means `Option<&T>` is the same size as `&T`), so wrapping fallible values costs nothing at runtime - the cost is purely at the call sites, which is exactly where you want the attention.